HK IMO Training 2020-2021
Problem Set 9
For 2 January 2020

Problem 33. All faces of the tetrahedron $ABCD$ are acute-angled triangles. The altitudes AK and AL of $\triangle ABC$ and $\triangle ABD$ are drawn where K and L are the feet of altitudes. It is known that the points C, K, L and D lie on the same circle. Prove that AB is perpendicular to CD.

Problem 34. Let $ABCD$ be a tetrahedron such that $AD \perp BC$ and $AC \perp BD$. Let E and F be the projections of point B on the lines AD and AC, respectively. Let M and N be the midpoints of AB and CD, respectively. Show that $MN \perp EF$.

Problem 35. Let $SABCD$ be a quadrangular pyramid whose base $ABCD$ is a convex quadrilateral. Suppose there is an inscribed sphere in the pyramid, and let P be the contact point of this sphere with the base $ABCD$. Prove that $\angle APB + \angle CPD = 180°$.

Problem 36. In a tetrahedron $ABCD$, $\triangle ACD$ and $\triangle BCD$ are acute-angled triangles. Let G and H be the centroid and the orthocentre of $\triangle BCD$. Let G' and H' be the centroid and the orthocentre of $\triangle ACD$. Given that HH' is perpendicular to the plane ACD, show that GG' is perpendicular to the plane BCD.